The Solutions

Question 1 (20 marks)

a)

The table below shows the pros and cons of both preventive and corrective modes.

Preventive	Corrective
No need for action	Need action (the changes are restricted by time limit). Benefit from automation
No violation in operating limits.	Operating limits may be temporarily violated.
No time limit	Time limit depending on the severity of the violation
High cost	Low cost

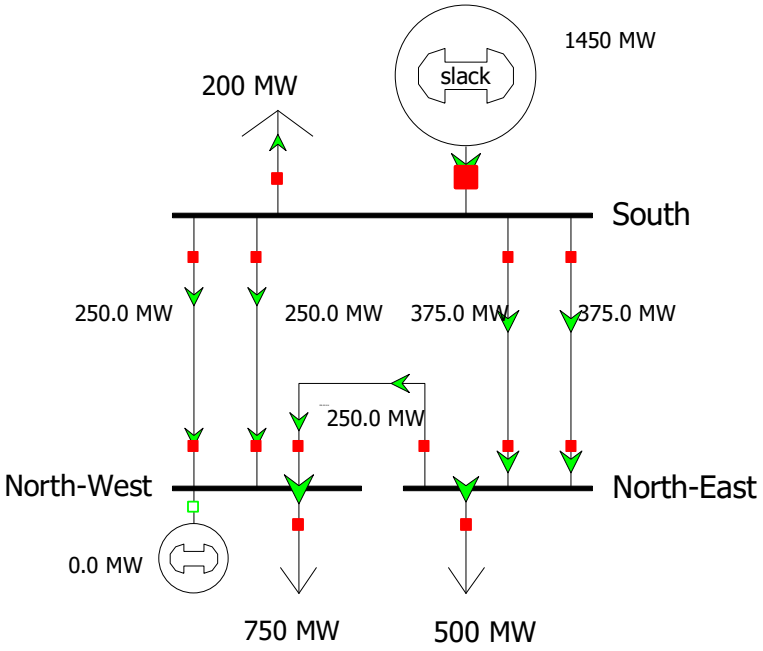
[1 per each row, upto 3 marks]

b)

i) Total load is 200 + 500 + 750 = 1450 MW. The ED solution: Gen at South produces 1450 MW and Gen at North-West produces 0 MW. Total cost is (1450*60 + 0*120) = £87,000/h. The flows can be calculated using superposition or dc load-flow.
[1 mark]

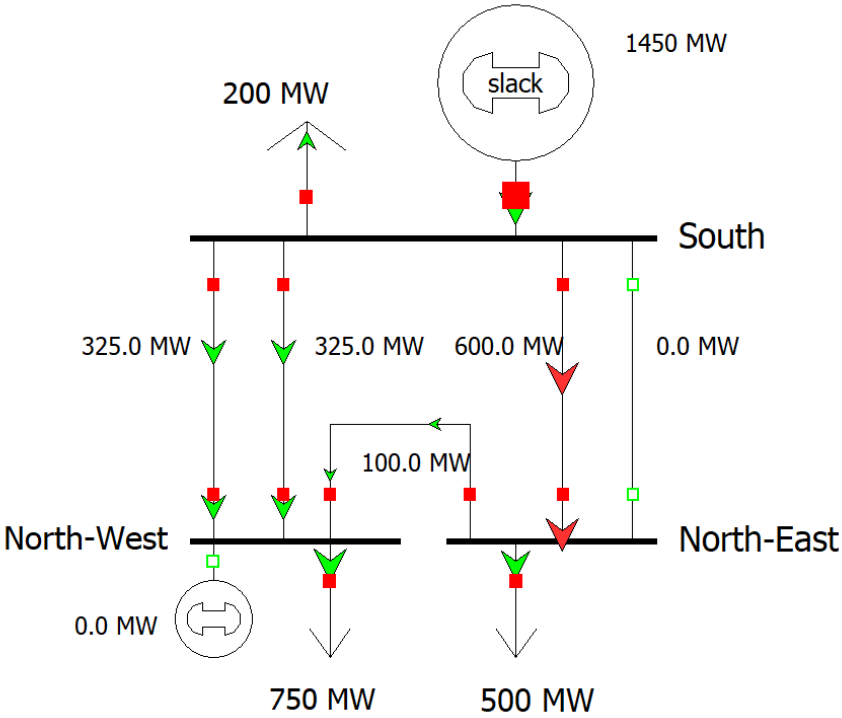
The flows for the intact system are as follows:

Model



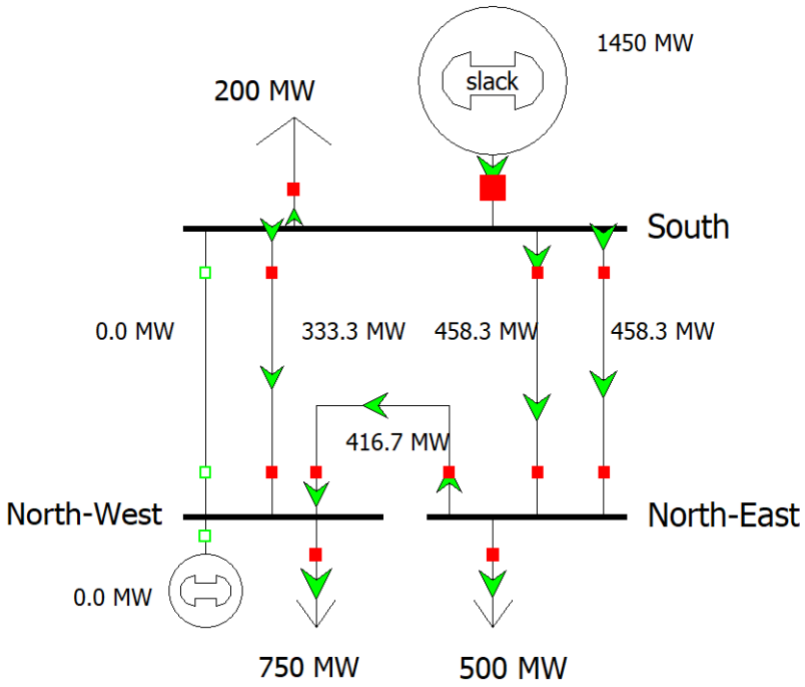
[1 mark]

ii) Flows when one circuit of South – North-East is out-of-service. Flows at the other circuit of the South – North-East corridor exceed the limit.



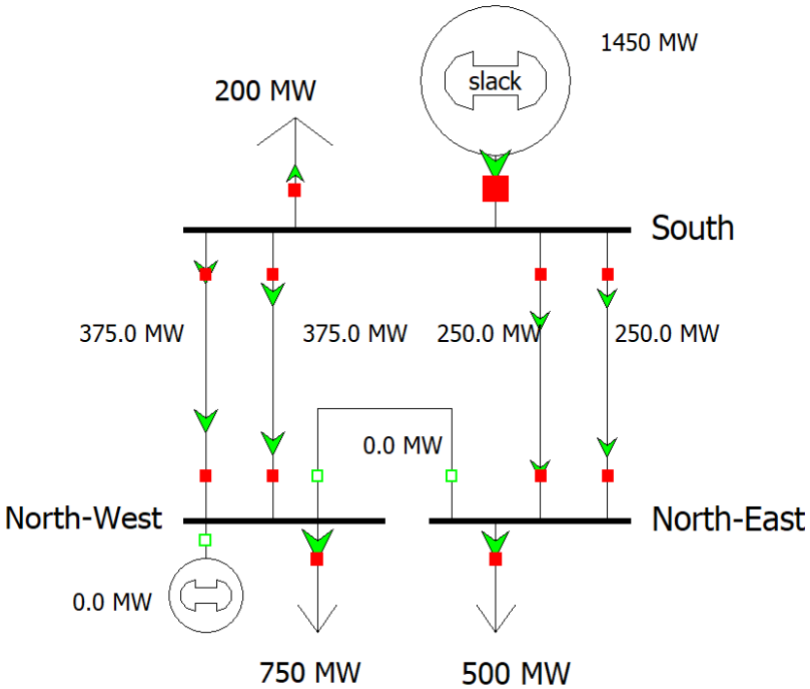
[1 mark]

Power lows when one circuit of South – North-West is out-of-service. No violation.



[1 mark]

Power flows when one circuit of North-East – North-West is out-of-service. No violation.



[1 mark]

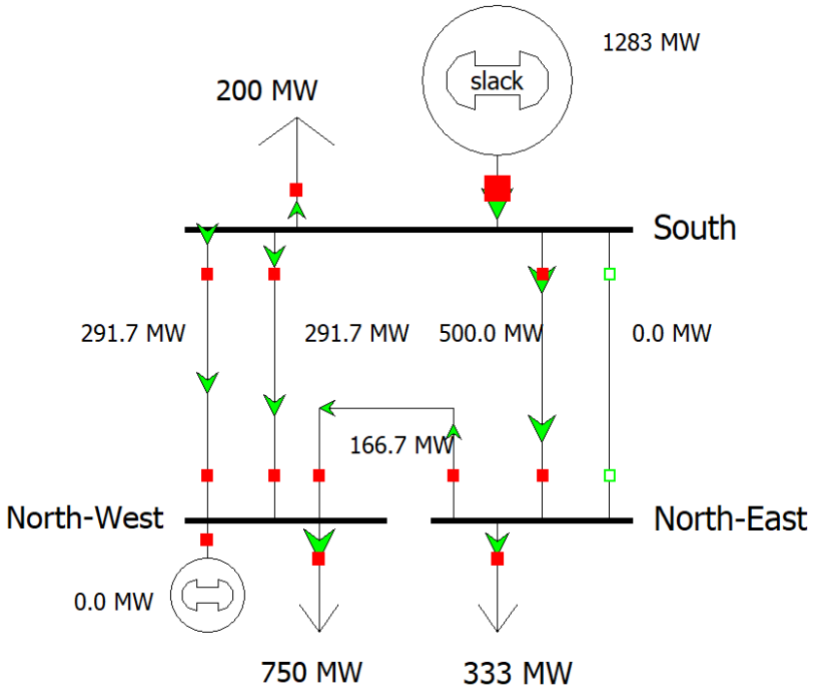
The critical contingency is the outage at the circuit between the South and North-East.

iii)

In order to reduce the flows from South and North-East from 600 MW to 500 MW, Demand Side Response of 166.67 MW at North-East is required. The remaining demand will be (500-166.67) = 333.33 MW.

[3 marks]

The flows are as follows:

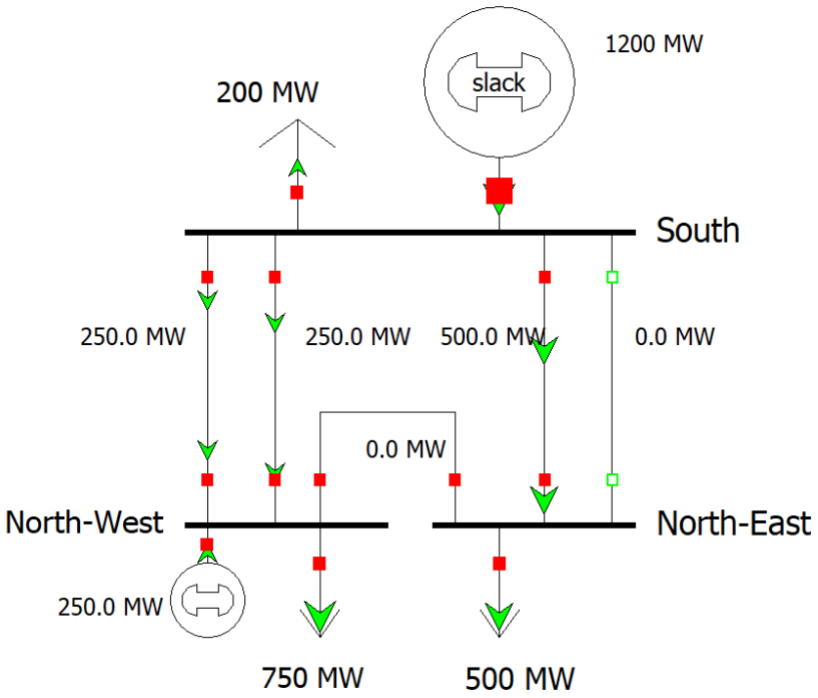


[3 marks]

iv)

To operate the system securely in the preventive mode, generator at North-West has to increase its production to 250 MW. The flows when one circuit of South-North-East is out of service are as follows:

Mock



[3 marks]

The operating cost is now (1200 x 60 + 250 x 120) = £102,000/h. Therefore, the cost of security is 102,000 – 87,000 = £15,000/h. Hence, the value of available Demand Side Response at North-East, providing post-contingency service in case of the loss of line between South and North-East, will be (15000/166.67)*100 = £9000/MW(rounded).

[3 marks]

Question 2 (20 marks)

a)
i)

The discharge time can be calculated using the formula: Discharge Time = Capacity / Discharge Rate

[1 mark]

Where:

Capacity = 100 MWh,

Maximum discharge rate = 20 MW.

Substituting the values:

Discharge Time = 100 MWh / 20 MW = 5 hours

Model Answers

Thus, the BESS can supply power for 5 hours at its maximum discharge rate before it needs to be recharged.

[1 mark]

ii)

The energy delivered to the grid is affected by the efficiency of the storage discharge power. The actual energy delivered is given by:

Energy Delivered = Capacity × Efficiency

[1 mark]

Substituting the values:

Energy Delivered = 100 MWh × 0.90 = 90 MWh

Thus, 90 MWh of energy will be delivered to the grid during the discharge period, accounting for the system's 90% efficiency.

[1 mark]

iii)

To calculate the percentage of the battery recharged, we first need to calculate how much energy is stored during the 5-hour charging period:

Energy Charged = Charging Power × Charging Time × Efficiency

Substituting the values:

Energy Charged = 15 MW × 5 hours × 0.90 = 67.5 MWh

[1 mark]

The percentage of the battery's capacity recharged is:

Percentage Recharged = Energy Charged / Total Capacity × 100

Substituting the values:

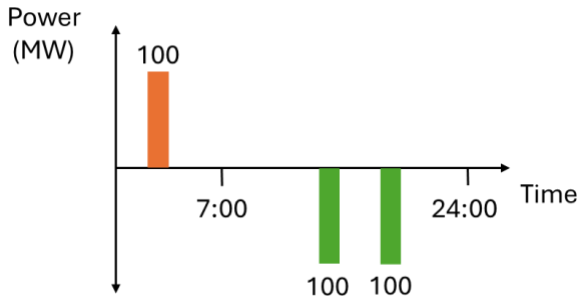
Percentage Recharged= 67.5 MWh / 100MWh × 100 = 67.5%

Thus, 67.5% of the battery's capacity will be recharged during the 5-hour charging period.

[1 mark]

b)

Model Answers

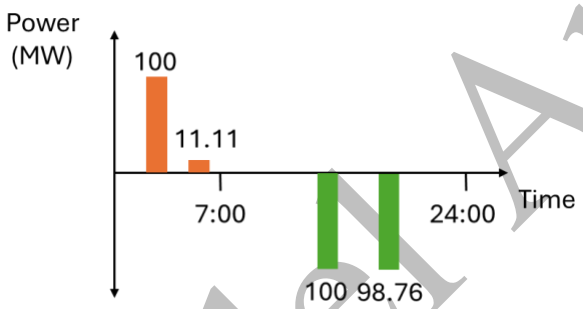


[2 marks]

Since the initial SoC is 50%, the battery will charge 100 MW at off-peak periods to reach its maximum SoC 100%, and then discharging 100 MW twice at peak periods to maximise the energy arbitrary.
Revenue = -100 × 5 + 200 × 10 = -500 + 2000 = 1500 £.

[2 marks]

c)



[2 marks]

When charging and discharging efficiencies are considered. The battery needs to charge more power to reach the maximum SoC 100%.

SoC_Charge1 = 50% + 100 MW × 0.9 / 200 MW = 95%

Charge2 = (1 – 95%) × 200 MW / 0.9 = 100/9 = 11.11 MW

[2 marks]

SoC_Discharge1 = 100% - 100MW/0.9 / 200 MW = 4/9 = 44.44%

Dicharge2 = 44.44% × 200 MW / 0.9 = 8000/81 = 98.76 MW

[2 marks]

Revenue = - (100 + 11.11) × 5 + (100 + 98.76) × 10 = - 555.55 + 1987.6 = 1432.05 £.

[2 marks]

Model Answers

d)

The annual income of the BESS is 1432.05 * 365 =522.7 k. If invested, there is a annual return of 522.7k- 400k = 122.7k. It is worth the investment.

[2 marks]

Question 3

a)

i)

1) The (Strong) Law of Large Numbers guarantees that $\lim_{n\to\infty} \frac{1}{n}\sum_{i=1}^{n} \mathrm{Im}(s_i) = E[\mathrm{Im}]$, which is used to estimate the true risk.

[1 mark]

2) The central limit theorem tells us that, for large n: $\overline{\mathrm{Im}} \xrightarrow{d} \mathcal{N}\left(\mu_{\mathrm{Im}}, \frac{\sigma^2_{\mathrm{Im}}}{n}\right)$, which is used to estimate the MC error distribution.

[1 mark]

3) Sample standard deviation is used to estimate the true standard deviation $\hat{\sigma}_{\mathrm{Im}} = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}\left(\mathrm{Im}(s_i) - \frac{1}{n}\sum_{j=1}^{n}\mathrm{Im}(s_j)\right)^2}$

[1 mark]

ii)

The coefficient of variation is defined as $\mathrm{CoV} = \mathrm{se}/\hat{r}$. The requirement is achieved when the standard error (σ') satisfies
σ'/110000=0.52%

[1 mark]

Hence, σ'= 572 and the required number of samples becomes
N'=N*(2093/572)^2=133889

[2 marks]

b)

Model Answers

Big data is high-volume, high-velocity and high-variety information made available through increased instrumentation in power systems.

[1 mark]

The primary sources of big data in power systems are:

- Phasor Measurement Units (PMUs)
- Asset monitoring data from various system assets e.g. transformers
- Smart meters
- Electricity market data

[2 marks for three or more examples, 1 mark for 1-2 examples]

c)

Traditionally, security assessment has been carried out at a limited, ad-hoc basis, largely based on empirical rules.

[1 mark]

Big data analytics can be applied to data gathered from phasor measurement units (PMUs) to carry out large-scale security assessment and improve situational awareness.

[1 mark]

In particular, historical PMU data contain information on what operating points the operator should anticipate in the future. Simulations of faults can be carried out to identify whether each anticipated operating point is stable or unstable. Machine learning techniques are then used to produce security rules describing the system's security boundary. In this way, the system operator can improve their situational awareness and very quickly identify potentially dangerous contingencies with respect to their current operating point.

[2 marks]

d)

i) Examples include device ownership, household income, daily routine and so on

[2 marks for three or more examples, 1 mark for 1-2 examples]

Model Answers

The exact information that can be extracted depends on the resolution data
[1 mark]

ii) List three from the below:
Pseudonymisation: Replacing identifiable features with consistent unique identifiers.
Aggregation: Aggregating consumption data across multiple periods of time (temporal) or households(spatial).
Differential Privacy: Carefully tuned noise addition to 'hide' individual contributions.
Homomorphic Encryption: Perform arithmetic operations (e.g., addition, multiplication) on encrypted data without having to first decrypt it.
User Demand Shaping: Altering actual consumption patterns using flexible assets to hide appliance characteristics.
Federated Learning: Distributed learning technique in which model parameters are shared but raw data is kept locally.
[2 marks for three or more examples, 1 mark for 1-2 examples]

The answer needs to cover the pros/cons of the listed three options based on the impact on utility of the smart meter data, the complexity of the implementation, and the compatibility with existing infrastructures.
[2 marks for three or more examples, 1 mark for 1-2 examples]

Question 4 (20 marks)

a)

The susceptance matrix can be found as

$$\mathbf{D} = \text{diag}\left(\frac{1}{x}\right) = \begin{bmatrix} 2.5 & 0 & 0 & 0 & 0 \\ 0 & 10 & 0 & 0 & 0 \\ 0 & 0 & 5 & 0 & 0 \\ 0 & 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 0 & 2 \end{bmatrix}$$

[2 marks]

The measurement matrix is then given as

$$\mathbf{H} = \mathbf{DA} = \begin{bmatrix} 2.5 & -2.5 & 0 & 0 \\ 10 & 0 & 0 & -10 \\ 0 & 5 & -5 & 0 \\ 0 & 5 & 0 & -5 \\ 0 & 0 & 2 & -2 \end{bmatrix}$$

[1 mark]

Model Answers

The rank of the measurement matrix is 3. This can be analytical computation or can be explained due to the existing of slack bus. The students can argue that the measurement matrix is rank three which equals to the number of unknown states or argue that the measurement matrix is full column rank after removing the first column.

[1 mark]

The system is observable as the rank of the measurement matrix equals to the number of non-slack bus.

[1 mark]

b)

The power injection can be found as

$$P_{inj} = A^T P_f = [5.5, 0.5, 1.3, -7.3]^T$$

[2 marks]

c)

After removing the first column of H,

$$H_{est} = \begin{bmatrix} -2.5 & 0 & 0 \\ 0 & 0 & -10 \\ 5 & -5 & 0 \\ 5 & 0 & -5 \\ 0 & 2 & -2 \end{bmatrix}$$

[1 mark]

$$\hat{x} = (H_{est}^T H_{est})^{-1} H_{est}^T P_f = [-0.2, -0.1, -0.5]^T$$

[1 mark]

Therefore, the total state can be

$$\hat{x} = [0, -0.2, -0.1, -0.5]^T$$

[1 mark]

d)

Is it impossible to only change $\mathbf{P_{13}}$ without being detected by the bad data detection (BDD).

[1 marks]

The students can argue that the attacked measurement and/or the attack vector is not in the column space of measurement matrix. The students can also calculate the residual of the attacked measurement, which is not equal to zero (note that this is not necessary)

[2 marks]

Model Answers

From $\mathbf{H_{est}}$, the state of bus 4 can be immediately calculated as $\frac{10}{-10} = -1.0$. Therefore, $\Delta\theta_4 = -0.5$ as only this state is changed. Consequently, the attack vector can be constructed as

$$a = H\Delta\theta = [0, 5, 0, 2.5, 1]^T$$

[2 marks]

e)

The existance of noise make the detection of FDIA harder as the detector has to distinguish between noise and FDIA.

[1 mark]

Because of this, it is impossible to guarantee the perfect detection without incurring false positives. When designing detector, one of the core consideration is the balance between detection rate and false positives.

[1 mark]

f)

By proactively changing the reactance of branches using D-FACTS devices, the attacker cannot know the latest power system configurations and the attack vector designed by the old measurement matrix can be detected by BDD under new measurement matrix.

[1 mark]

Because only the state of bus 4 is changed, reactance of transmission line that is connected to bus 4 needs to be changed. Therefore, the options can be transmission line 1-4, 2-4, or 3-4.

[2 marks]

Model Answers